

G.A.T.E.S. — Gate 1

Execution validity, evidence provenance,
and measured Agent Laboratory impact

40 / 40

required values
delivered

FINAL PROFESSOR SUBMISSION

10 gated + 10 ungated artifacts · 362 passing tests · DeepSeek V4 Flash
MLR-Bench comparator · combined Claude + Codex analysis · portable verification bundle

Gate 1 improved evidence fidelity—not scientific judgment

40/40 vs 0/40

required values delivered

+100 percentage points

4/4 vs 0/4

required values in papers

gated writer received the registry

96.6% vs 0%

paper-claim traceability

direct registry + checkable derivations

Accept Gate 1

for deterministic execution validity, typed result capture, and value-layer provenance.

Paper quality did not materially improve. Gate 1 does not verify scientific interpretation or manuscript compliance.

A deterministic trust boundary between code and claims



The verdict is model-free. Optional model scanning and fix prose cannot admit or reject an execution.

One common configuration; one intentional evidence-window stressor

10 + 10

execution artifacts

same count in both arms

a0409cf...

shared YAML hash

exact config identity

>1,000 chars

logging before final metrics

deliberately crosses the legacy boundary

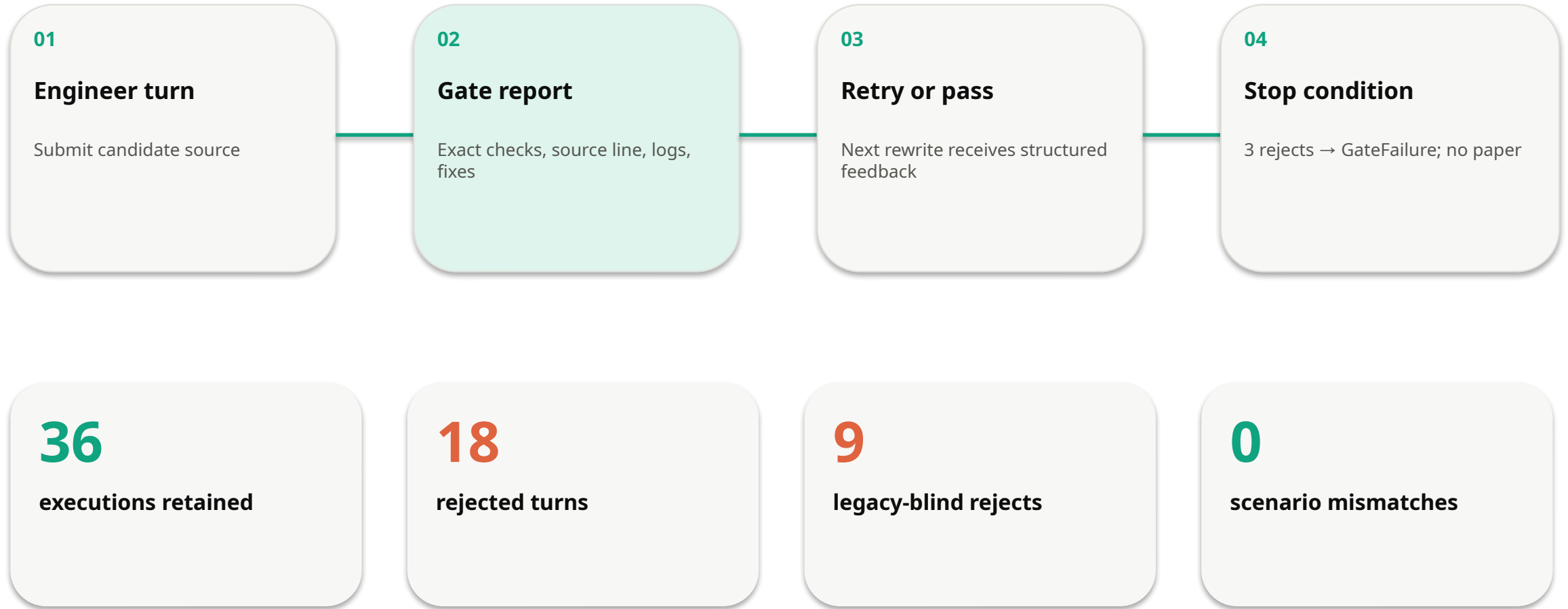
REQUIRED CONTRACT

`eff.acc_at_25 · eff.acc_at_100`
`eff.efficiency_ratio · eff.train_s`

Both writers were told: if evidence is absent, say so—never estimate or reconstruct a result.

Arm switch: GATES_GATE1=on/off. No credential was stored in YAML, argv, source, or logs.

A failed verdict becomes bounded repair context



The loop returns evidence the agent can locate and act on

ACTUAL ATTEMPT 2 REPORT — ABRIDGED

GATE 1 — EXECUTION VALIDITY: FAIL (attempt 2 of 3)

```
[exec.exit_code_zero] exited 1  
[exec.no_uncaught_exception] NameError: name 'correct' is not defined  
[results.contract_present] no results were declared – the run crashed before any  
record_result() call completed
```

```
warning: no seed was declared  
stdout: 3,420 bytes captured in full  
source hash: 0f4e60778e3e...  
line 8: test_acc = correct / total
```

Why it mattered

The crash marker landed at character 3,442.

The legacy detector only received the first 1,000 characters and would have accepted the run.

Gate 1 retained the full log, rejected the exception, and returned three concrete fixes.

22 checks: 13 block; 9 diagnose and document

13 BLOCKING

STATIC

syntax_valid
no_unbound_names
no_banned_calls
contract_not_shadowed

EXECUTION

completed_within_budget
exit_code_zero
no_uncaught_exception
code_identity
clean_namespace

RESULTS

contract_present
expected_keys_present
values_computed
values_finite

6 WARN + 3 INFO

DIAGNOSTIC

no_swallowed_traceback
no_error_signals
model_error_signals
seed_recorded
single_observation
non_degenerate

AUDIT

output.untruncated
env.provenance
report.fixes_grounded

Full gated campaign: every blocker passed 10/10. Nine attempts warned on changing duplicate timing values; final attempt was clean.

The code often computed results; only Gate 1 delivered a citable evidence bundle

GATE 1 ON

10/10 PASS

- 40/40 required values recorded and delivered
- 10 reports + 10 citable registries
- 4,099–6,962 bytes of stdout retained in full

`workflow.log`: GATE 1 – EXECUTION VALIDITY: PASS

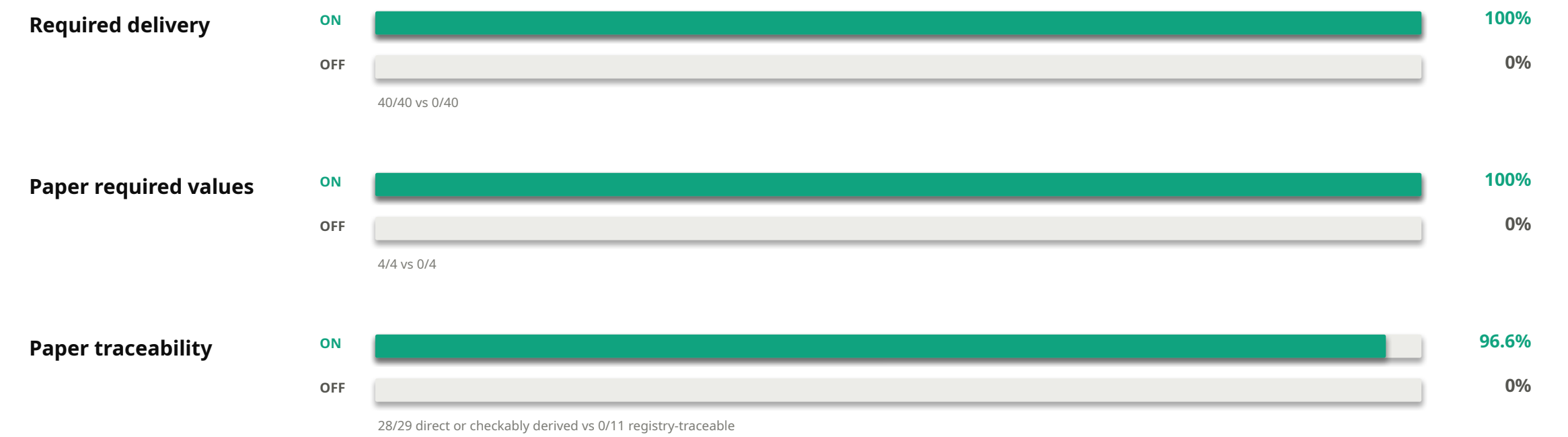
GATE 1 OFF

9/10 ACCEPTED

- 36/40 required values recorded; 0/40 delivered
- One visible syntax failure
- 19,945 characters captured; 1,000 handed on

`workflow.log`: gate 1 BYPASSED – 1,000 handed on

The evidence channel is the causal comparison



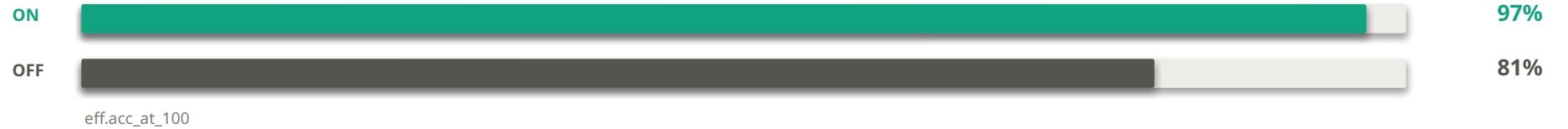
+100 percentage points in required-result delivery on the same artifacts

Raw accuracy differs, but it is not a Gate effect

100 labels



400 labels



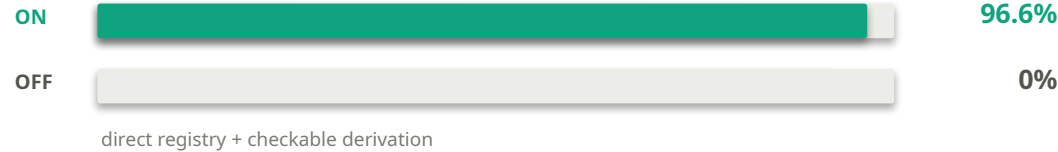
Efficiency ratio



The agents sampled different generators and training schedules. Do not attribute +11.5 or +16.0 accuracy points to Gate 1.

Evidence quality improved; reviewer scores barely moved

Traceability



Writer reward



Reviewer mean



OUT OF SCOPE

SCIENTIFIC INTERPRETATION

The gated paper still added an unjustified two-point scaling exponent and violated its report contract.

Later-gate work.

Complete output can still be unsound

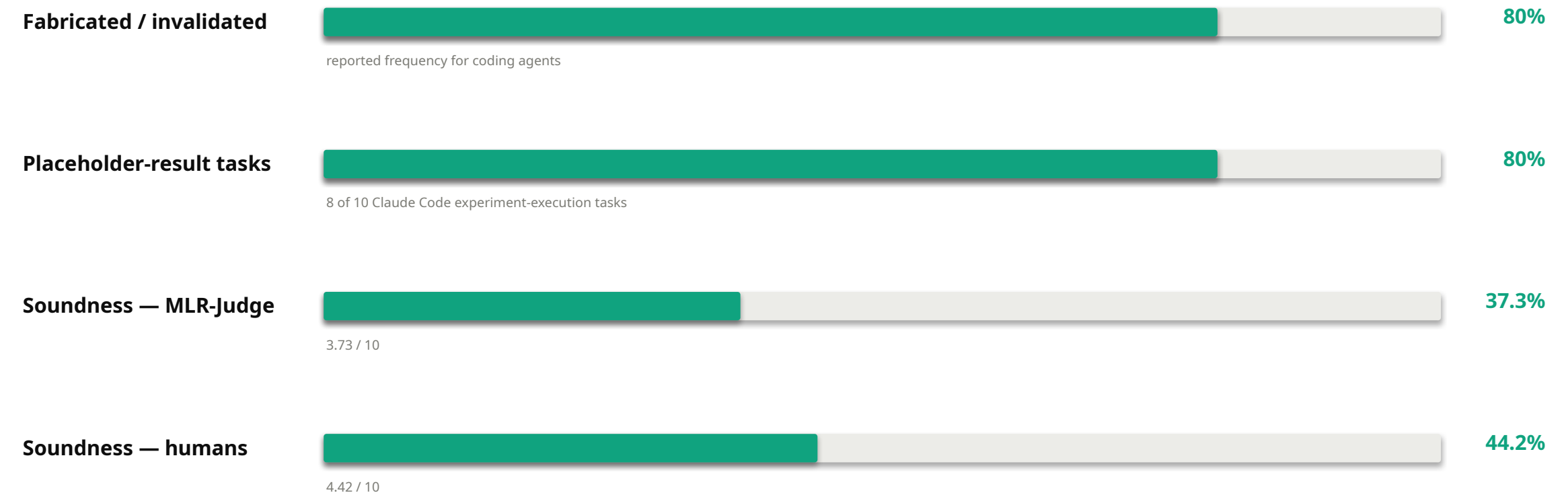


Table 5 overall scores: Claude Code 4.95 / 10 · Codex 4.95 / 10 · acceptance threshold 6.0

Gate 1 closes one evidence failure mechanism



Not an MLR-Bench submission. The 80% published failure rate and this campaign’s percentages are contextual—not one shared accuracy scale.

What changed—and what did not

01

+100 pp

Evidence delivery

Every required value reached the writer with a citable run origin.

02

−0.030

Reviewer-score gap

Generic reviewer means were nearly identical; no paper-quality effect is claimed.

03

not causal

Raw accuracy

Different sampled experiments explain the 0.89/0.97 vs 0.775/0.81 values.

Gate 1 makes measurements auditable. It does not make the science correct.

Accept Gate 1; continue the validation stack

ACCEPT GATE 1

- Deterministic execution validity
- Typed result capture and complete logs
- Source → command → log → value provenance
- Bounded repair feedback
- 286 + 76 tests passing

Next: Gate 3, strict result schemas, real sandbox, ≥10 paired MLR-Bench tasks.

EVIDENCE PACKAGE

Report

GATE1_FINAL_REPORT.pdf

Feedback report + full logs

verification/evidence/feedback_loop_example/

Workflow logs

reports/finalized-report-and-results/verification/logs/

Benchmark

reports/finalized-report-and-results/verification/
benchmark/MLR_Bench.pdf

Everything needed to verify the result

SUBMIT

Final report

GATE1_FINAL_REPORT.pdf

16-slide presentation

GATE1_FINAL_PRESENTATION.pptx

Portable preview

GATE1_FINAL_PRESENTATION.pdf

VERIFY

- Gated + completed ungated workflow logs
- Both generated papers, LaTeX, and figures
- All 20 completed-arm execution artifacts
- Gate reports, registries, feedback-loop repeats
- Shared YAML, manifests, source reports, MLR-Bench
- SHA-256 checksum manifest

verification/ • **SHA256SUMS.txt** • **SUBMISSION_MANIFEST.json**